

# Diamond Cut 32 Bootcamp

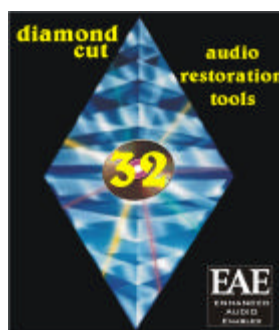
EnhancedAudio.com

Getting Started Series  
Volume 1

## Welcome To The World's Best Audio Restoration

EnhancedAudio.com provides the world's best tools for enhancing today's audio. But we also provide the world's best tools for restoring yesterday's audio. Our **Diamond Cut 32** software is the best affordable noise reduction/audio enhancement software on the market today. You get more tools, more control, and power that most people never dreamed possible for restoring old vinyl, tape or any other older recording techniques.

Of course, just because we provide you with the best tools in the industry, doesn't necessarily mean that you'll be an old pro the instant that you fire up the software for the first time. These tools are powerful, and with that power comes a certain learning curve that you'll have to traverse before you feel totally confident with the software. To that end, we feel that it is our responsibility to provide you with the best tutorial knowledge that we can... No matter what your level of audio understanding. Bootcamp is our way of taking you through important basic training. It will give you ideas and helpful hints so that, when you finish, you'll understand how we go about doing things. Of course, to be totally productive with the software, you'll also want to make full use of our extensive online help files, the PDF manual, which you can secure at [www.enhancedaudio.com/tutorial.htm](http://www.enhancedaudio.com/tutorial.htm), and the knowledge database at [Enhancedaudio.com](http://Enhancedaudio.com) as well.



Incredible Affordable Power

### Now Drop and Give Us 20!

Just kidding.

This tutorial does assume that you have a working sound card hooked to speakers in your computer system. It also assumes that you understand that (no matter what you paid for your computer) most likely, you are using some sort of sound solution that came bundled with your computer. Keep in mind that, no matter what you paid, you probably have the equivalent of a \$12 sound card in your machine. It will generate quite a lot of noise when you record and play back. The noise that is added when you record will make its way to your CD recordings. Let your ears be the judge. If you're not happy with what you're hearing...visit the [www.enhancedaudio.com](http://www.enhancedaudio.com) store and think about a sound card investment. We won't sell you more than you need and there are quality cards starting at \$49!

## Let's Get Started (Wired for Sound)

Your sound card is just like any other audio device that you're familiar with. To record music to the hard drive of your computer is almost exactly the same process that you've used before to record your records to cassette tape. Most sound cards use an 1/8" to RCA left and right Sony cable for inputs and outputs. If you need to get one or two of these cables (they are standard Walkman cables) you can find them at any Radio Shack or electronics-type store. You'll simply plug the left and right RCA cables to the output of your audio gear, and the 1/8" plug into the Input of your sound card. The output probably contains an 1/8" mini plug that runs to your computer's speakers. Keep in mind that, if you are planning on plugging your turntable directly to your sound card, you'll need some sort of Preamp to boost the signal enough so that you can get solid recording levels in your recording. If you don't have a "Tape Out" in your stereo, or just can't physically move your whole stereo to your computer, go to the [EnhancedAudio.com](http://EnhancedAudio.com) store. We have very nice preamps starting at \$89 for Diamond Cut users...as well as a special that you can get Diamond Cut 32 and a Preamp for only \$129. You will not need a preamp for most cassette tape or CD player outputs.

Now that you're wired for sound, let's try a recording. We'll assume installation went well, if not, call us!

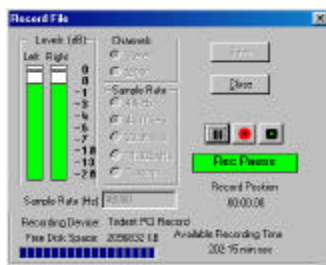
### Inside BootCamp

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### Highlights of Diamond Cut 32

- Remove Clicks, Pops, Hiss, Hum, Buzz, Surface noise, etc.
- Prep for CD Mastering
- Listen to changes "while you make them in real-time!"

Provides a full set of audio enhancement tools to reinvigorate old recordings, without compromising the original feel of the material.



Diamond Cut 32's Recording Screen

## Let's Try A Recording

Keep in mind that Diamond Cut 32 can work with any existing 16 bit Wave file (.WAV) which is the standard format that Windows PC computers use for digital audio. In order to record a new file:

1. Make sure you've got the outputs of the device you want to record plugged to the inputs of your sound card
2. To set the record levels, you can either go to Edit/Record File or hit the record button (4th Icon, Red Circle w/black dot)
3. Let's check to see if we have levels. Click on the Pause button (two vertical lines beside the Record button) and start your tape deck or phonograph playing. You should see green level meters jump to life. You can now adjust the volume by opening the sound card's mixer (small speaker icon in the tray beside your clock in Windows) going to Options/Properties/Recording and adjusting the Line Level input up or down. (Do not use your Microphone Input...as they are very noisy) If you can't get levels at all, you should check under Edit/Device I/O in Diamond Cut 32 and make sure your sound card is selected for Input. If this is okay, you may want return to the Recording mixer (small yellow speaker again) and make sure the Line Input is not muted. If none of these work, you have a sound card problem and should contact your manufacturer for new drivers or suggestions.
4. Try and keep your recording levels in the high green but try to avoid the red. Don't worry about not getting maximum levels on recording...the final step in most Noise reduction jobs is a Gain Normalize, which will give you maximum levels available without distortion. *Note: If you clip on recording, you'll have very little head room left for applying any Equalizer boost, etc. without distorting.*

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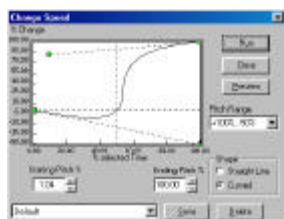
5. Click on the Record button and start recording. Try a small file at first....until you get the hang of it. *Note: If you've recorded a wave file and you hear it, but can't see the whole wave form, go to the Preferences Menu and adjust the Display Length to a larger size. This is a feature, provided so that you don't have to wait for long screen redraws.*

## The Source and Destination Windows

Once you've recorded a minute or so, stop recording and hit the SAVE button. After you name the file, Diamond Cut will draw the waveform in the top window. This is called the Source window. Whenever a tool is applied, the top window is affected and the results show in the bottom window. Many times in Noise Reduction you will try many different things to arrive at the desired final product. This way of doing things allows you the option to keep several renditions and then choose the best. Keep in mind, that if you apply a tool to the Source window's wave file and you want to keep that and apply something else to it, you must first change the Destination file to a Source file so that the tool can be applied to your newest file. To do this, just go to File/Make Destination Source and your Destination file will open in a new window as the Source.

This is a very good reason why Plug-in noise reduction just doesn't work very well. In noise reduction, you need to be able to compare and listen to several layers of before and after...this isn't very graceful from the front end of a sequencer or other audio products. Diamond Cut 32 saves your takes and allows you to audition them.

## Tools, Tools, Tools



Typical Diamond Cut 32 Tool

Diamond Cut 32 features a huge toolbox of both Noise Reduction and Audio Enhancement tools. It can be a little intimidating at first, but we'll review some of the big ones here and take you step by step through several small restoration projects. Even though these tools offer a wide range of functionality, after you begin to work with them one by one, you'll discover that they all have a similar feel.

Most of the Diamond Cut tools have real-time auditioning...that simply means that you can start playing the wave file, apply the tool and make parameter adjustments while you listen to the way it affects the wave file. Unlike many audio restoration tools, Diamond Cut 32 allows you to "taste-test" the tools before you let the computer go away and chug through the math. To start the file playing...just hit the Preview Button.

Most of the tools present some sort of graph with colored boxes that can be moved to affect the sound while you listen. Don't worry...as you work with these tools, their functions will become more apparent.

You weren't an expert with Word the first time you used it either...were you?

## Removing Clicks and Pops on a Vinyl LP

The number one concern of most vinyl restoration customers is the removal of the nasty clicks and pops that find their way onto precious recordings. Let's start there. We'll be using the Impulse Filter. Go to Filter/Impulse Noise. The Impulse Noise Filter is also good for old radio broadcasts for removing static clicks.

1. Click on "Vinyl LP" An "x" will appear in the box adjacent to the feature.
2. Set the Threshold Control to its minimum setting of "1"
3. Set the Tracking Control somewhere in the 25 to 30 range to start with.
4. Set the Size Control to a value somewhere in the range of 10 to 15 samples.

*Note: You can also accomplish steps 1-4 by going to the Presets drop-down and choosing "Vinyl LP Declick Start Point"*

5. Run the filter or use Preview mode, and determine if the wave file is being adequately de-clicked. If it is not, lower the Tracking Control. If it is de-clicking, but producing distortion on the sibilant sounds, then raise the control. Continue this process until a good balance is established of minimum sibilant distortion and minimum click feed-through. If it is de-clicking, but leaving a bit of an artifact behind, increase the value of the Size Control.
6. This is a good time to experiment with the different settings by moving the sliders up and down to hear how they affect the sound.
7. When you determine the best setting of the control for your particular wave file, click on Run Filter.

When the filter has completed its operation, the results will appear in the "Destination" workspace.

**Note:** Unlike all of the other **Diamond Cut 32** controls, Vinyl L.P. mode can not be turned on or off "live" in Preview Mode.

**Note #2 :** If you are restoring 78 RPM records, you'll need to consult the manual for appropriate steps!



The Impulse (Declick) Filter

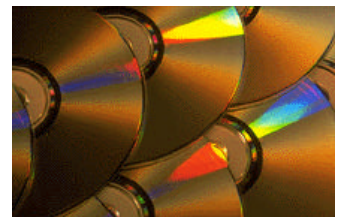
## Removing Hiss and Other Constants

For those of you restoring music or other material from Tape, it will be likely that you'll need a little visit with our Continuous Noise Filter. You can access this filter via the Filter/Continuous Noise Menu. The idea of this filter is to sample small segments of noise within your source file...fingerprint that noise, and then remove it from the entire piece. This is especially effective in Hiss and Surface Noise reduction.

1. Before we enter the Continuous Noise Filter, we'll want to take a small sample of just noise. This can usually be found at the intro of a cassette tape or the Lead In or Exit Groove on a record. Highlight about 1 second of material. To select an area, just left click at the beginning of the desired part and drag to the end...it should now be highlighted. *Note: Using more than 1 second of material will likely just confuse the filter and not provide the desired results.* Now open the Continuous Noise Filter and click on Sample Noise. After some computation, you'll see a graph appear that contains red and blue lines that represent (red) your sample and (blue) Diamond Cut 32's idea where to begin the process of reducing the noise. You'll also see 10 touch points which represent areas that you can boost or attenuate to fine tune the filter.
2. Now that we've got the desired sample...you'll want to activate the entire wave file for playback. Just double click in the main window to select your entire recording for playback. Now when you click the Preview button, you hear the file with some noise reduction in place.
3. Now you've got to make a judgment about your file. Is your major disturbance happening in the upper frequencies (high end, like hiss)? Think of the scale as left (low) and right (high), and then move one of the blue touch points. Unlike a normal equalizer, up increases filtering and down decreases filtering. In the case of Hiss, start at the third touch point to the right and move it down slightly. Now you'll hear the noise return...edge it up until the noise goes away again. You can repeat as necessary with the three touch points on the right until you can "dial in" the filter.

**There's an excellent tutorial file that you can download at [www.enhancedaudio.com/tutorial.htm](http://www.enhancedaudio.com/tutorial.htm). It is a wave file that matches the tutorial in the software under Help/Contents/Demo file example restoration procedure.**

*"Don't get frustrated. The Continuous Noise Filter is the trickiest filter in Diamond Cut 32 and you will need to practice to perfect your technique."*



We'll check out some CD Prep techniques later in the tutorial!



The Dynamics Processor

## Enhancing Audio With Diamond Cut 32

In most cases, noise removal is only 1/2 the game. To truly return your original recording to its former shape, you'll need to use a series of very cool enhancement tools that only Diamond Cut 32 offers. For Enhancement help, you'll want to investigate the Paragraphic Equalizer, Dynamics Processor, Reverb, and Virtual Valve Amplifier (see below)

These tools are like candy, you want to eat enough to get that sugar buzz, but you don't want to overeat, or you'll end up with an unhappy stomach...or in this case...unhappy ears.

There's really no official way to

use these effects, better to do a little experimentation to see what you prefer.

Reverb is a pretty nice effect that adds some depth to older recordings...but don't fall in love with it. It is probably also the most overused effect...remember the "Candy analogy". The Paragraphic Equalizer is also a very cool device that, when used in moderation can really improve your material.

Diamond Cut 32 includes over 200 factory presets...when you're new to the program...try them. Just go to the bottom of each effect and you'll see a white box with the word

"Default" in it. This is the preset area. Presets are a group of settings that we've prepared at the factory for you. They may not be exactly the effect you're looking for, but they may be a very good starting point. After you find a preset that gets you in the "ballpark" just experiment until you find the one that pleases your ear and then save it with a name that you can use in the future.

To save your original preset, just click on the save button.

*"Diamond Cut 32 can take you from a barely audible old recording to a fully restored treasure that's ready for CD storage in a few easy steps"*

## CD Prep With Diamond Cut 32

Diamond Cut 32 does not directly interface with your CD Recorder. We leave that job to the software the specializes in it. We'd rather our programmers are writing cool new algorithms than spending their time writing drivers for the newest hardware. Diamond Cut will, however get your files totally prepared for CD Recording. This is a very easy process.

Though some of this is repetitive from earlier tutorials, there is some new stuff that could help you:

1. Record your files at 44.1 16 bit stereo. You can record one side of an album or cassette at the same time

2. Using Edit/Cut, you can remove excess time at the beginning and end of the file.

3. Use the Impulse Noise Filter and Continuous Noise Filter discussed earlier, to remove clicks, pops, hiss and surface noise. You can also use the enhancement tools to spice up your sound.

4. Click on the CD Prep Menu.

5. Click on Find and Mark Silent Passages

6. Check to be sure that the markers have located themselves correctly between "cuts". Adjust them if necessary.

9. Click on "Quantize for CD Audio"

10. Click on "Chop File Into Pieces"

11. Repeat the above process for side 2.

Using your CD-R software, create a recording list from the individual wave files which you have just created and then "burn that ROM!" (Make your CD)



The Virtual Valve Amplifier

## The Virtual Valve Amplifier

By far, one of the coolest effects offered by Diamond Cut 32 is the Virtual Valve Amplifier. The VVA simulates actual vacuum tubes and can be wonderful for breathing new life into your old recordings.

Many audio aficionados have complained over the years at the lack of warmth provided by

compact discs. Now you can add anything from simple warmth to full fledged guitar distortion to your files.

Just like the other effects, you're probably better off starting out by experimenting with the factory presets just to see what each tube can do. Try moving the sliders and using different

tubes until you find the desired effect.

Triode Tube Preamp 3 is very cool.



## Diamond Cut Millennium...the best gets better!

From a tool standpoint, Diamond Cut 32 is hard to beat. It has the largest assortment of noise reduction and audio enhancement tools on the planet...so how do you make it better? You make it easier to use, more powerful and more flexible. The result...Diamond Cut Millennium.

With Diamond Cut Millennium, we've added:

Batch File Processing- Process hundreds of files with one click!

Multi-Filter- Now you can apply a series of filters and enhancers simultaneously. You simply drag the tools you desire, program them the way you want them and then apply them to one or more of your audio files. These two features alone are especially handy for more experienced users.

We've also added:

- 8, 16, 20, and 24 Bit File Support
- Up To 96 kHz Sampling Rate Support
- Variable Resolution in the Continuous Noise Filter
- Audio Spectrum Enhancement in the Dynamic Noise Filter
- Slot Filter to Enhance Forensic Applications
- Stereo 180 Degree Phase Inversion
- X-Y Output Display for Plotting Vector Waveforms
- Improved Impulse Noise Filter with HQ Mode
- Stereo Channel Blender/Crossover Filter and Effect
- Improved Reverb

- Printable 50/60 Hz Turntable Stroboscope Discs
- Even More Factory Presets
- A suite of 5 Forensics Audio Filters, including an Adaptive Filter
- Improved Resolution Spectrum Analyzer with Peak Readout
- Drop Markers "On-The-Fly" feature
- Independent left and right channel Manual Interpolate function
- MP3 to Wavefile Decoder
- New Keyboard Accelerators
- List Price \$199

**Registered Diamond Cut 32 Users Upgrade for only \$99**



## Diamond Cut LIVE...Real time Support!

The big brother in the Diamond Cut Family... Diamond Cut LIVE is designed for advanced users in the fields of Forensics or Broadcast, who need to use the advanced tools of Millennium, but who need those tools applied in real time.

With Diamond Cut LIVE, you process as fast as your material plays. You simply plug your source material into the input of your sound card, set up a series

of filters and enhancers on your computer screen using the Multi-Filter and then start your source material playing. The resulting "cleaned" material feeds out of your sound card's output in Real time.

This product is ideal for users who have large amounts of material to process, but without the time necessary to record to the hard drive of the CPU, then process the audio, then play back

to some device.

LIVE saves time and money for those with limits on both.

It has a list price of \$995 but is available to registered Millennium users for only \$399.



## Still Using That Bundled Sound Card?

You've created such an advanced environment for recording. You've got the fastest computer, a great 12X CD recorder, and now, you've got the world's best Noise Restoration software. So why do you bring this whole system to its knees by using the lousy \$12 sound card that came bundled with your PC? Here's a hint...if it says Gold, Platinum, LIVE, Blaster or anything like it, no matter what you paid for it,

most likely, it's not very good. You probably didn't care about such things when you were just playing games or listening to the various boops and beeps that your computer generates. But now, you're going to be making CDs that will be listened to on your expensive Car CD or Home Stereo. Now you're going to notice the weak link in your system. Have no fear, good sound cards are not as expensive

as you may think (start at \$49) Why not pay a little visit to the EnhancedAudio.com store and equip yourself with the proper tools to do the job? We offer products that give you true CD recording at a price that anyone can afford.

Go ahead...your ears deserve it!

*"No matter what you paid for your computer...you most likely have the equivalent of a \$12 noisy sound card as the heart of your audio."*

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### Enhancing The Internet Audio Experience



Our goal at EnhancedAudio.com has always been to provide you with cutting edge technology that anyone can afford.

EnhancedAudio.com is currently tied to the [www.tracertek.com](http://www.tracertek.com) store. Tracer provides a laundry list of products that have great quality and a fair price. If you need Sound Cards, USB sound products, Multi-track hardware, Multi-track software, Studio quality hardware, or just have a question about audio in general, check out this site.

Here are some of the products that may interest you:

**SAS**— If you want an excellent all purpose sound card that has great specs, a wonderful MIDI synthesizer, sampling capabilities, and a design that is specific to CD recording, then check this out. Only \$159 [www.tracertek.com/tracer.htm](http://www.tracertek.com/tracer.htm)

**Studio 8**— Use your current stereo sound card and have your own 8 track digital recording studio for less than \$3 per track. Studio 8 offers 8 tracks of recording, full level and pan automation, and real time effects. For \$19.95, you can really see if you have the chops to release your own musical CD. Easy learning curve! Also available with 16 tracks for only \$179

**CD Recorder III**— Make your own original audio CDs with the click of a button. CD Recorder III integrates with your CD and makes wonderful audio CDs without the complication of some CD Recording software. Only \$49

**The WaMi Box**— Now notebook computer owners needn't be out in the cold! The WaMi Box offers a full high quality audio solution with a portable attitude...it includes full MIDI synth, full 20 bit analog and digital ins and outs, Mic preamp...everything you need to make music. Only \$399

**For More info on all products, go to [www.enhancedaudio.com](http://www.enhancedaudio.com) and click on the Store button.**

## What Does This Logo Mean?

It means that the web site or product you are using cared enough to offer you the best possible audio experience available. EnhancedAudio.com has devoted itself to enhancing your overall audio experience...whether it be on the Internet or some sort of portable device. Due to the massive compression required for standards like MP3 or streaming audio, you pay a certain price in the quality of the sound you hear for the convenience of quicker downloads...unless of course, your system is Enhanced Audio Enabled. EnhancedAudio.com deals with today's audio and develops algorithms designed to improve them.

If you don't see this logo, ask the person who provides the MP3 files or the web site, or the streaming audio you're listening to why they don't have the Enhanced Audio abilities.

Your ears will thank you.



## Questions?

We realize that it's not possible to cover every possible question that you may have in one 6 page document. Please feel free to send any questions you may have to: [questions@enhancedaudio.com](mailto:questions@enhancedaudio.com)

Wanna check out the site? Go to [www.enhancedaudio.com](http://www.enhancedaudio.com) and look around. The site is always growing, so if you haven't been there for awhile, you may want to wander on back. If you want a full PDF of the Diamond Cut 32 Manual, go to [www.enhancedaudio.com/tutorial.htm](http://www.enhancedaudio.com/tutorial.htm)



Your One Stop CD Making Source